

Sharp binary-quartic rigidity and the optimal order of tensor error bounds

Research note, 7 October 2026. Complete written proofs with a partial Lean formalization of the scalar quartic certificate. Literature comparison is preliminary; no priority assertion is made.

1. Statements and conventions

Let T be a real fully symmetric order- p tensor on $\mathbb{R}^2$, with $p \geq 3$. Write t_k for an entry having exactly k indices equal to 2. Our tensor norm and associated polynomial are

$$\|T\|_F^2 = \sum_{k=0}^p \binom{p}{k} t_k^2, \quad f_T(x, y) = \sum_{k=0}^p \binom{p}{k} t_k x^{p-k} y^k.$$

The orthogonally decomposable class is

$$\mathcal{D}_{2,p} = \{au^{\otimes p} + bv^{\otimes p} : (u, v) \text{ an orthonormal basis, } a, b \in \mathbb{R}\}.$$

Zero, repeated and negative weights are allowed. For every **ordered** index tuple $\alpha \in \{1, 2\}^{p-2}$, let $(X_\alpha)_{ij} = T_{\alpha, i, j}$, and set

$$R_p(T)^2 = \sum_{\alpha, \beta \in \{1, 2\}^{p-2}} \|[X_\alpha, X_\beta]\|_F^2.$$

Thus R_p has degree two in T ; it is already a square root of a sum of squared commutator norms. This convention matters for all constants below. Let $d_p(T) = \text{dist}_F(T, \mathcal{D}_{2,p})$.

Theorem 1 (binary all-orders bound). For every $p \geq 3$,

$$d_p(T) \leq \frac{\sqrt{p}}{2} \sqrt{R_p(T)}.$$

In particular, $R_p(T) = 0$ if and only if $T \in \mathcal{D}_{2,p}$.

Theorem 2 (sharp quartic bound and equality). For $p = 4$,

$$d_4(T) \leq \frac{3^{1/4}}{\sqrt{2}} \sqrt{R_4(T)}.$$

The constant is best possible. For $R_4(T) > 0$, equality holds exactly for nonzero real scalar multiples and orthogonal coordinate transforms of the tensor T_* with coefficients

$$(t_0, t_1, t_2, t_3, t_4) = (4, \sqrt{3}, 0, \sqrt{3}, 4).$$

For $R_4(T) = 0$, both sides vanish.

Theorem 3 (order growth in fixed dimension). For every $p \geq 5$, the tensor S_p with

$$t_0 = t_p = 1, \quad t_1 = t_{p-1} = p^{-1/2}, \quad t_k = 0 \quad (2 \leq k \leq p-2)$$

satisfies

$$\|S_p\|_F^2 = 4, \quad d_p(S_p)^2 = 2, \quad R_p(S_p) = \frac{4\sqrt{2(p-1)}}{p}.$$

Consequently, if C_p is the best constant in a binary estimate $d_p(T) \leq C_p \sqrt{R_p(T)}$, then

$$\frac{\sqrt{p}}{\sqrt{2}[2(p-1)]^{1/4}} \leq C_p \leq \frac{\sqrt{p}}{2} \quad (p \geq 5).$$

There is no constant uniform in tensor order, even in dimension two.

Theorem 4 (matching upper growth). For every $p \geq 3$,

$$C_p \leq \min\{\sqrt{p}/2, (3p/4)^{1/4}\}.$$

In particular $C_p = \Theta(p^{1/4})$ as $p \rightarrow \infty$. An improved explicit bound is available: define

$$c_p = \begin{cases} (p-1)/(p-2), & p \text{ even}, \\ p/(p-1), & p \text{ odd}. \end{cases}$$

For $p \geq 9$,

$$C_p^2 \leq \frac{p}{4\sqrt{p/(4c_p) - 1}}.$$

Thus the normalized constants satisfy

$$2^{-3/4} \leq \liminf_{p \rightarrow \infty} \frac{C_p}{p^{1/4}} \leq \limsup_{p \rightarrow \infty} \frac{C_p}{p^{1/4}} \leq 2^{-1/2}.$$

The leading constant, and whether the normalized constants converge, remain undetermined.

These statements are independent of entropy inequalities or any claimed resolution in the OpenAI collection. They refine our earlier all-dimensional tensor note, whose substantially larger dimension-dependent constants remain valid. The earlier exact cubic constant is $\sqrt{3}/2$; Theorem 2 concerns the next order, with a different equality configuration.

2. Distance and a two-dimensional Gram matrix

Lemma 4 (projection formula).

$$d_p(T)^2 = \|T\|_F^2 - \max_{(u,v) \text{ orthonormal}} (f_T(u)^2 + f_T(v)^2).$$

For a fixed basis, the best weights are $a = f_T(u)$, $b = f_T(v)$.

Proof. The tensors $u^{\otimes p}$ and $v^{\otimes p}$ are orthonormal in the Frobenius inner product. Minimize the two separate quadratic expressions in a, b . A maximizing basis exists by compactness of $O(2)$ and continuity. This also proves existence of a nearest element of $\mathcal{D}_{2,p}$.

Let $w_k = \binom{p-2}{k}$, $0 \leq k \leq p-2$, and define

$$z_k = \begin{pmatrix} t_k - t_{k+2} \\ 2t_{k+1} \end{pmatrix}, \quad G(T) = \sum_{k=0}^{p-2} w_k z_k z_k^\top.$$

Lemma 5 (exact residual identity and change of basis).

$$R_p(T)^2 = \det G(T).$$

Under orthogonal coordinate changes, G is conjugated by an orthogonal two-dimensional matrix. Every rotation of this trace-free coordinate plane can be obtained by a rotation of the underlying tensor coordinates.

Proof. A contraction with k indices equal to 2 is

$$X_k = \begin{pmatrix} t_k & t_{k+1} \\ t_{k+1} & t_{k+2} \end{pmatrix}$$

and appears w_k times. The commutator $[X_k, X_l]$ has off-diagonal entries $c_{kl}, -c_{kl}$, where

$$c_{kl} = (t_k - t_{k+2})t_{l+1} - (t_l - t_{l+2})t_{k+1} = \frac{1}{2} \det(z_k, z_l).$$

Expanding the 2×2 determinant gives

$$\det G = \frac{1}{2} \sum_{k,l} w_k w_l \det(z_k, z_l)^2 = 2 \sum_{k,l} w_k w_l c_{kl}^2 = R_p(T)^2.$$

For the basis-change assertion, first index the contraction family by all ordered tuples. A rotation Q acts on its label space by the orthogonal matrix $Q^{\otimes(p-2)}$, and on each matrix by conjugation with Q . The first operation leaves the sum of the trace-free outer products unchanged. On a matrix X , the vector $(X_{11} - X_{22}, 2X_{12})$ transforms orthogonally under conjugation. A coordinate rotation through θ rotates this vector through 2θ , up to the harmless sign convention for whether the new basis is active or passive. Therefore every plane rotation is available. This proves the assertions, including reflections.

Proof of Theorem 1. Choose tensor coordinates making $G_{22} = \lambda_{\min}(G)$ and $G_{12} = 0$. Use the approximation retaining only t_0, t_p in this basis. Its squared error is

$$E = \sum_{k=1}^{p-1} \binom{p}{k} t_k^2 \leq p \sum_{k=1}^{p-1} \binom{p-2}{k-1} t_k^2 = \frac{p}{4} G_{22}.$$

Indeed, $\binom{p}{k} / \binom{p-2}{k-1} = p(p-1)/(k(p-k)) \leq p$. As G is positive semidefinite, $G_{22} = \lambda_{\min} \leq \sqrt{\det G} = R_p(T)$. Lemma 4 gives $d_p^2 \leq E$, proving the bound. If $R_p = 0$, the same argument gives an exact decomposition. Conversely, every tensor in $\mathcal{D}_{2,p}$ has diagonal contractions in its decomposing basis, so its residual vanishes.

3. The sharp quartic constant

We prove $4d_4(T)^4 \leq 3R_4(T)^2$. The zero tensor is immediate. For a nonzero tensor, choose a maximizing basis in Lemma 4, and write

$$L = t_0 + t_4, \quad D = t_0 - t_4, \quad h = t_1 + t_3, \quad j = t_1 - t_3, \quad t = t_2, \quad N = L^2 + D^2.$$

Here $N > 0$: a nonzero homogeneous polynomial is nonzero at some point of the unit circle, and hence the maximum projection energy is positive. The squared distance in this basis is

$$E = d_4(T)^2 = 2(h^2 + j^2) + 6t^2.$$

3.1 An exact criterion for a maximizing basis

On the circle, write

$$f_T(\cos \theta, \sin \theta) = a \cos 4\theta + b \sin 4\theta + c \cos 2\theta + d \sin 2\theta + e,$$

where direct expansion gives

$$a = (L - 6t)/8, \quad b = j/2, \quad c = D/2, \quad d = h, \quad e = 3(L + 2t)/8.$$

The projection energy for a rotated orthonormal basis is

$$2(e + a \cos 4\theta + b \sin 4\theta)^2 + c^2 + d^2 + (c^2 - d^2) \cos 4\theta + 2cd \sin 4\theta.$$

Apart from its constant part, this is

$$q(z) = 2(az_1 + bz_2)^2 + \ell_1 z_1 + \ell_2 z_2, \quad z_1^2 + z_2^2 = 1,$$

where $\ell_1 = 4ea + c^2 - d^2$, $\ell_2 = 4eb + 2cd$. Stationarity at our chosen basis $z = (1, 0)$ says $\ell_2 = -4ab$, equivalently

$$Lj + Dh = 0.$$

Moreover, using $z_2^2 = 1 - z_1^2$, we have the identity

$$q(1, 0) - q(z) = (1 - z_1)[\ell_1 + 2(a^2 - b^2)(1 + z_1) + 4abz_2].$$

The minimum of the bracket on the unit circle is $\ell_1 + 2(a^2 - b^2) - 2(a^2 + b^2) = \ell_1 - 4b^2$. Consequently, subject to stationarity, $(1, 0)$ is a global maximizing point if and only if $\ell_1 \geq 4b^2$. Necessity follows also when the minimum is at $(1, 0)$: a negative value there remains negative at nearby points, for which $1 - z_1 > 0$.

Because $N > 0$, stationarity lets us write

$$h = sL, \quad j = -sD$$

for some real s . Put

$$u = L/\sqrt{N}, \quad \tau = t/\sqrt{N}, \quad \sigma = s^2, \quad v = u + 6\tau, \quad k = (4 - v^2)/16.$$

Then the global maximizing condition is exactly

$$|u| \leq 1, \quad |v| \leq 2, \quad 0 \leq \sigma \leq k.$$

To check the last assertion without division ambiguities, calculate

$$16(\ell_1 - 4b^2) = 4N - (L + 6t)^2 - 16s^2N.$$

3.2 A complete nonnegative polynomial certificate

The Gram entries of Lemma 5 in this basis are

$$G_{11} = \frac{1}{2}(L - 2t)^2 + \frac{1}{2}D^2 + 2j^2, \quad G_{22} = 2(h^2 + j^2) + 8t^2, \quad G_{12} = 2tj.$$

Thus

$$\frac{R_4^2}{N^2} = \left(\frac{1}{2} - 2u\tau + 2\tau^2 + 2\sigma(1 - u^2)\right)(2\sigma + 8\tau^2) - 4\tau^2\sigma(1 - u^2), \quad \frac{E}{N} = 2\sigma + 6\tau^2.$$

Let $P = (3R_4^2 - 4E^2)/N^2$. Expansion yields

$$P = 3\sigma + 12\tau^2 - 12u\tau\sigma - 48u\tau^3 - (48 + 36u^2)\tau^2\sigma - (4 + 12u^2)\sigma^2 - 96\tau^4, \quad \tau = (v - u)/6.$$

For fixed u, v this is a strictly concave quadratic in σ . Its endpoint values have the following exact factorizations:

$$P(u, v, 0) = 12\tau^2[\frac{4}{9}(u - v/4)^2 + 1 - v^2/4],$$

$$1728P(u, v, k) = 16(4u - v)^2(u - v)^2 + 27(1 - u^2)(4 - v^2)Q(u, v),$$

where

$$Q(u, v) = 4u^2 - 8uv + v^2 + 8 = (2u - v)^2 + 4(2 - uv) \geq 0.$$

The last inequality follows from $|u| \leq 1, |v| \leq 2$. Both endpoints are nonnegative. For completeness, the following division-free interpolation identity is a direct polynomial check:

$$kP(u, v, \sigma) = (k - \sigma)P(u, v, 0) + \sigma P(u, v, k) + (4 + 12u^2)k\sigma(k - \sigma).$$

If $k = 0$, necessarily $\sigma = 0$. If $k > 0$, every term on the right is nonnegative. Hence $P \geq 0$, proving the claimed quartic inequality. These identities, the nonnegativity implication for **all real** u, v, σ in the displayed domain, and the link to the residual expression are checked in `formal/QuarticCertificate.lean` without unproved placeholders.

3.3 All equality cases

Suppose $R_4 > 0$ and $P = 0$. If $k > 0$ and $0 < \sigma < k$, the last term of the interpolation identity is positive, which is impossible. Thus an endpoint must occur.

At $\sigma = 0$, either $\tau = 0$, yielding an odeco tensor and $R_4 = 0$, or

$$v = \pm 2, \quad u = v/4, \quad \tau = v/8.$$

At $\sigma = k > 0$, we have $|v| < 2$, so $Q > 0$. The second endpoint formula forces $|u| = 1$. Its first term then forces $v = u$, since $v = 4u$ is outside the domain. Therefore

$$u = \pm 1, \quad \tau = 0, \quad \sigma = 3/16.$$

The latter configuration has $D = t_2 = 0$ and

$$t_0 = t_4 = \lambda, \quad t_1 = t_3 = \pm \frac{\sqrt{3}}{4}\lambda, \quad \lambda \neq 0.$$

Reflection of one coordinate changes the sign of t_1, t_3 , so these are exactly the scalar multiples of T_* up to orthogonal changes.

In the first nonzero-residual configuration $h = j = 0$, and, after an overall sign and possible interchange of the axes, the coefficients are

$$\frac{\sqrt{N}}{4}(1 + \sqrt{3}), \quad 0, \quad \frac{\sqrt{N}}{4}, \quad 0, \quad \frac{\sqrt{N}}{4}(1 - \sqrt{3}).$$

Rotating T_* by $\pi/4$ gives $(2+2\sqrt{3}, 0, 2, 0, 2-2\sqrt{3})$; scaling by $\sqrt{N}/8$ gives precisely this configuration. Thus it belongs to the same orbit.

Finally, T_* has $\|T_*\|_F^2 = 56$, $G = \text{diag}(32, 24)$, and projection maximum 32. The maximizing criterion above certifies that the standard axes attain this maximum: $a = 1, b = c = 0, d = 2\sqrt{3}, e = 3$ give $\ell_1 = 0 = 4b^2$. Hence

$$d_4(T_*)^2 = 24, \quad R_4(T_*)^2 = 768, \quad \frac{d_4(T_*)^2}{R_4(T_*)} = \frac{\sqrt{3}}{2}.$$

This proves sharpness, as well as sufficiency of every listed equality case, by homogeneity and orthogonal invariance. The zero-residual equality cases were already classified by Theorem 1.

4. An infinite family forcing order growth

Proof of Theorem 3. The norm formula immediately gives $\|S_p\|_F^2 = 2 + 2p/p = 4$. Since $p \geq 5$, the four specified entries do not overlap. Direct substitution in Lemma 5 gives

$$G(S_p) = \text{diag}(4(p-1)/p, 8/p),$$

so the claimed residual follows.

It remains to prove the maximum projection energy is exactly 2, rather than infer this from numerical maximization. Write $x^2 + y^2 = 1$, $a = x^2$, $b = y^2$, so $a + b = 1$, and set

$$B = a^p + b^p + p(ab^{p-1} + ba^{p-1}).$$

For odd $p \geq 5$, expanding $f_{S_p}(x, y)^2 + f_{S_p}(-y, x)^2$ gives

$$2B + 4\sqrt{p}(ab)^{(p-1)/2}.$$

The binomial expansion of $(a + b)^p = 1$ contains the four terms in B . The two central terms, which are not among those four, sum to

$$\binom{p}{(p-1)/2}(ab)^{(p-1)/2}(a + b) = \binom{p}{(p-1)/2}(ab)^{(p-1)/2}.$$

All remaining terms are nonnegative. Thus

$$B \leq 1 - \binom{p}{(p-1)/2}(ab)^{(p-1)/2}.$$

The central coefficient is at least p , and $p \geq 2\sqrt{p}$ for $p \geq 5$, proving that the projection energy is at most 2.

For even $p \geq 6$, the corresponding exact expansion is

$$2B + 4(p+1)(ab)^{p/2}.$$

The omitted central binomial term gives $B \leq 1 - \binom{p}{p/2}(ab)^{p/2}$. Here

$$\binom{p}{p/2} \geq \binom{p}{2} = p(p-1)/2 \geq 2(p+1) \quad (p \geq 6),$$

again proving the required upper bound. The coefficient comparisons used in both cases follow from the successive ratio $\binom{p}{k+1}/\binom{p}{k} = (p-k)/(k+1)$ up to the middle. The final even-case comparison is $(p^2 - 5p - 4)/2 \geq 0$, true at $p = 6$ and increasing thereafter.

These formulas hold for all signs of x, y : the cross powers are $(xy)^{p-1}$ in the odd case and $(xy)^p$ in the even case, both even powers. Every orthonormal basis is covered, since changing the sign of a basis vector does not change its squared evaluation. The standard axes have projection energy 2. Lemma 4 therefore gives $d_p(S_p)^2 = 4 - 2 = 2$. Taking its ratio to the residual proves the asserted lower bound for C_p and its $p^{1/4}$ asymptotic order.

5. Matching upper growth by trace contraction

Define the symmetric order- $(p-2)$ tensor $(\text{Tr } T)_\alpha = T_{\alpha,1,1} + T_{\alpha,2,2}$. The binomial identity $\binom{p}{k} = \binom{p-2}{k} + 2\binom{p-2}{k-1} + \binom{p-2}{k-2}$, with out-of-range coefficients zero, gives

$$\text{tr } G(T) = 2\|T\|_F^2 - \|\text{Tr } T\|_F^2.$$

This can also be read entry by entry from $(a-c)^2 + 4b^2 = 2(a^2 + 2b^2 + c^2) - (a+c)^2$.

For a symmetric binary cubic with coefficients (a, b, c, d) ,

$$4(a^2 + 3b^2 + 3c^2 + d^2) - 3[(a+c)^2 + (b+d)^2] = (a-3c)^2 + (d-3b)^2 \geq 0.$$

Fix any ordered $(p-3)$ -tuple and apply this bound to the remaining three indices of T . Summing over all such tuples proves, for every $p \geq 3$,

$$\|\text{Tr } T\|_F^2 \leq \frac{4}{3}\|T\|_F^2, \quad \|T\|_F^2 \leq \frac{3}{2} \text{tr } G(T).$$

The sum on the left is exactly the norm of the trace tensor, because its remaining one index together with the fixed tuple ranges over all ordered $(p-2)$ -tuples; no multiplicities are discarded.

Let $A = \lambda_{\max}(G)$ and $B = \lambda_{\min}(G)$. In the basis used in Theorem 1 the squared distance $E = d_p(T)^2$ satisfies both

$$E \leq \frac{p}{4}B, \quad E \leq \|T\|_F^2 \leq \frac{3}{2}(A+B).$$

Since $A \geq B \geq 0$, multiplying these two nonnegative bounds yields

$$E^2 \leq \frac{3p}{8}B(A+B) \leq \frac{3p}{4}AB = \frac{3p}{4}R_p(T)^2.$$

This proves the first assertion of Theorem 4. The scalar multiplication step, cubic trace bound and entrywise trace identity are also Lean-checked. Together with Theorem 3, this determines the optimal growth order.

5.1 A sharper trace norm and explicit constant

We supply a complete derivation of the sharper trace estimate

$$\|\text{Tr } T\|_F^2 \leq \sigma_p \|T\|_F^2, \quad \sigma_p = \frac{4\lfloor p^2/4 \rfloor}{p(p-1)}.$$

Complexify the tensor space with its Hermitian Frobenius inner product. The vectors $e_+ = (e_1 + ie_2)/\sqrt{2}$ and $e_- = (e_1 - ie_2)/\sqrt{2}$ are Hermitian orthonormal. Their bilinear Euclidean contractions, which define the complex extension of the real trace, satisfy

$$e_+ \cdot e_+ = e_- \cdot e_- = 0, \quad e_+ \cdot e_- = 1.$$

Let $F_{p,r}$ be the average over all distinct permutations of the tensor with r copies of e_+ and $p-r$ copies of e_- . Its $\binom{p}{r}$ summands are Hermitian orthonormal, so $\|F_{p,r}\|_F^2 = 1/\binom{p}{r}$, and different r give orthogonal tensors. Tracing the first two slots keeps precisely the permutations where one slot is e_+ and the other e_- . Their proportion is $2r(p-r)/(p(p-1))$. The remaining slots have the uniform symmetrization, so

$$\text{Tr } F_{p,r} = \frac{2r(p-r)}{p(p-1)} F_{p-2,r-1} \quad (1 \leq r \leq p-1),$$

with zero output for $r=0, p$. On the normalized orthonormal basis $\sqrt{\binom{p}{r}} F_{p,r}$, the squared nonzero singular values are

$$\left(\frac{2r(p-r)}{p(p-1)}\right)^2 \frac{\binom{p}{r}}{\binom{p-2}{r-1}} = \frac{4r(p-r)}{p(p-1)}.$$

Distinct input indices map to distinct orthogonal output indices. Thus the largest squared singular value is σ_p , proving the estimate for all complex tensors and hence all real tensors. The norm is attained over the real space as well: for even p , $F_{p,p/2}$ is real; for odd p , take the real sum of the two conjugate middle basis vectors, whose singular values agree. This attainment is not needed for the upper bound.

It follows that $\|T\|_F^2 \leq c_p(A+B)$ with $c_p = (2 - \sigma_p)^{-1}$ as stated. If $B=0$, the distance is zero by Theorem 1. Otherwise write $x = \sqrt{A/B} \geq 1$. The two distance bounds now give

$$\frac{d_p(T)^2}{R_p(T)} \leq \min\{p/(4x), c_p(x + 1/x)\}.$$

For $p \geq 9$, we have $p \geq 8c_p$ (check $p=9$ directly and use $c_p \leq 9/8$ for $p \geq 10$). The decreasing and increasing functions cross at $x_* = \sqrt{p/(4c_p) - 1} \geq 1$. For $x \leq x_*$ use the second bound, and for $x \geq x_*$ use the first. Their common value at the crossing is $p/(4x_*)$, proving the sharper assertion of Theorem 4. Since $c_p \rightarrow 1$, its asymptotic upper constant is $2^{-1/2}$.

6. Verification, dependence and open questions

The proofs above do not depend on finite sampling. The standard-library exact checker independently expands the polynomial certificate, computes the sharp quartic tensor in $\mathbb{Q}(\sqrt{3})$ using the complete ordered contraction convention, checks rational orthogonal transformations, and replays the order-growth identities and binomial comparisons at specified finite orders. The finite order replay is a diagnostic for formulas; the infinite family is proved in Section 4. Optimized Python execution also passes, since validation does not use removable `assert` statements.

Lean formalizes the nine scalar results listed in the formal directory, including the domain-wide inequality used in Section 3.2. It does not formalize the tensor definitions, angular reduction, equality classification or the infinite binomial argument. Its exported axioms are recorded; `sorryAx` and new postulated axioms are absent. This is model-conducted verification, not external human peer review or whole-paper formalization.

The original qualitative odeco characterization is part of the established tensor literature; we do not claim it as a first discovery. The relevant comparison includes Boralevi–Draisma–Horobet–Robeva, *Orthogonal and unitary tensor decomposition from an algebraic perspective*, Israel Journal of Mathematics 222 (2017), 223–260, DOI 10.1007/s11856-017-1588-6. Our preliminary benchmark comparison records exactly which sources were inspected; it does not establish priority.

Two principal quantitative gaps remain. First, what is the sharp asymptotic leading constant in $C_p = \Theta(p^{1/4})$, and can all-order extremizers be classified? Second, in higher dimension can the earlier polynomial-dimensional upper bound be reduced toward the known $m^{1/4}$ direct-sum obstruction while retaining the complete commutator residual and arbitrary weights? Both require further theorems.